Supplementary material

S1. Data processing

WorldCover built-up components connected by eight-neighbour adjacency and covering at least 1 km² were buffered by 2 km, clipped to the administrative unit and gridded at 200 m in UTM 49N (Zhengzhou) or 50N (Beijing). Cells with water fraction above 0.5 or DSM coverage below 0.8 were removed. The ADM2 selectors are Zhengzhoushi (17275852B14866353147502) and Beijingshi (17275852B40522342172572), from geoBoundaries gbOpen CHN commit 9469f09592ced973a3448cf66b6100b741b64c0d. Terrain processing extended 10 km beyond each study area.

The source windows are 19 July 2021 19:03 to 4 August 16:58 for Zhengzhou, and 30 July 2023 14:50 to 11 August 21:42 for Beijing (UTC+8). Of 2,131 / 784 source records, mutually exclusive filtering removed 529 / 339 outside the retained grid, 734 / 138 failing the all-place-names-unique geocoding rule, and 77 / 11 subsequent records within an already retained cell, leaving 791 / 296 cells. Unique geocoding identifies one candidate location, not independently verified positional accuracy.

Copernicus GLO-30 DGED 2024_1 elevations were aggregated by valid-area weights and filled with float64 Priority-Flood; equal-height queue entries use row–column order. D8 maximizes positive descent divided by neighbour distance. Ties follow east, southeast, south, southwest, west, northwest, north, northeast. Drainable flats use an integer mask combining distance away from higher ground with twice the distance toward lower outlets; routing follows strictly lower mask values without perturbing elevation. Unresolved flats trigger an error.

Slope uses Horn's 3×3 operator; relative elevation and relief use centred 5×5 windows. Flow accumulation includes the cell itself and is log-transformed. Wetness is $\ln((a + 1\ \mathrm{m})/\tan(\max(\beta, 0.5°)))$, with specific catchment area a and slope β.

IMERG rates are multiplied by 0.5 h and assigned from the nearest native pixel. The 72 h antecedent window ends at the record-window start above and selects 144 half-hour midpoints in [start−72 h, start); event totals include midpoints from start through end. Maximum 6 h rainfall uses 12 consecutive selected intervals, advanced by 30 min. Complete intervals are integrated without endpoint clipping; incomplete sums are missing and maxima use finite complete windows. Event totals can include rain after individual reports.

Land-cover fractions require at least 50% valid coverage. OSM snapshots use each record-window end; distances are capped at 10 km. WorldPop counts for 2021 / 2023 are aggregated conserving totals and converted to log density.

Table S1. Environmental variables.

Group	Variable	Unit
Terrain	Elevation	m
Terrain	Slope	degree
Terrain	Relative elevation within 1 km	m
Terrain	Local relief within 1 km	m
Terrain	Log flow accumulation	dimensionless
Terrain	Topographic wetness index	dimensionless
Rainfall	Record-window total	mm
Rainfall	Maximum 6 h total	mm
Rainfall	72 h antecedent total	mm
Land cover	Built-up fraction	fraction
Land cover	Water fraction	fraction
Land cover	Vegetation fraction	fraction
Land cover	Cropland fraction	fraction
Roads and waterways	Road length density	km km^{-2}
Roads and waterways	Waterway length density	km km^{-2}
Roads and waterways	Distance to nearest mapped waterway (OSM)	m
Land cover	Distance to mapped water (land-cover mask)	m
Population	Log population density	ln(1 + persons km^{-2})

S2. Fitting and evaluation details

Spatial blocks are ordered by decreasing recorded-positive count, then cell count, row and column. Greedy assignment minimizes squared normalized imbalance, weighting cell imbalance by 0.2; ties use current positive count, cell count and fold ID. Sampling uses sorted cell IDs and PCG64DXSM, with SHA-256 namespace-derived seeds. Exact namespaces, seed integers and refit seeds accompany the code snapshot.

Training samples comprise 100 revealed positives and 2,000 unlabelled cells; unrevealed recorded positives can enter the latter. The two sampled classes receive equal total loss weight. Median imputation and standardisation use training samples only; zero standard deviations become one.

Table S2. Candidate settings. Each hidden layer uses the stated width.

Setting	Hidden width	Dropout	Learning rate	Weight decay
Smaller	32 per hidden layer	0.2	0.001	0.0001
Larger	64 per hidden layer	0.4	0.0005	0.0005

Both networks use ReLU, dropout and a final linear binary output. Fixed propagation precedes a two-hidden-layer network. GraphSAGE sums separate transformations of own and incoming-neighbour representations. AdamW runs for at most 250 epochs with patience 30 and no learning-rate scheduler. The median selected inner best epoch sets refit duration. All 350 main selections, their three inner epochs and actual refit durations are supplied. The separate attribute series reuses D8-selected settings and durations across graph references; 30 all-positive-budget records lack archived durations, which remain unavailable.

Seed-only comparisons use all three unordered own-model refit pairs and both signs within each city-fold-revelation unit. Pair contributions retain full rankings and the same support groups. Fixed-pair city means average five revelations per fold, then five folds. A randomized reference independently selects one of six ordered pairs per unit, retaining common pair choices across groups, over 10,000 draws (seed 20261029). Its central 95% range is descriptive: it uses one pair per unit, whereas the graph contrast averages three refits per model, so it is not a variance-matched null test. Each graph contrast is also recalculated after omitting one refit slot. Selected settings, durations, training samples and preprocessing are fixed across seeds within each model. Cross-model slots need not share the same integer seed when selected settings differ. Aggregate arithmetic, exact-tie handling and reconstruction of all archived structural fold effects were checked.

Let D(F,b) be D8-minus-own sampled AP for attributes F and label budget b. The budget interaction is [D(F,100)−D(18,100)]−[D(F,all)−D(18,all)]. Direct removal effects instead subtract each model's own 18-attribute AP. Foldwise effects are supplied; no archived intervals were available for direct model-level removal effects.

Sampled AP averages three fixed banks containing M recorded positives and 10M background cells. Its recorded-positive fraction is 9.091%; whole-city recorded-positive shares are 0.482% in Beijing and 3.145% in Zhengzhou, not flood-occurrence rates. Constant-score AP is M/(M+wU). Full-domain reweighting changes w, retaining every candidate.

For recall capacity 0.1N, the score at rank ceil(0.1N) defines the boundary. Cells above it enter; each tied cell receives inclusion weight (0.1N−n_above)/n_tied. Capacity can be fractional even with a single boundary cell. Recall divides expected retrieved recorded positives by M.

AP contributions retain the complete candidate ranking; empty groups contribute zero. Group mean precision changes divide each fold's contribution by its positive share before averaging. Rank means require group presence in every fold; rank and mean-precision ranges require at least five distinct positives per group in every fold. Equal-fold means differ from pooled rankings of independently fitted models.

S3. Synthetic implementation

Each grid contains 96×48 cells. Training uses rows 0–31, validation 36–59 and testing 64–95; edges remain within partitions. Features use training-row standardisation. Logistic fitting uses L-BFGS-B for at most 300 iterations, with mean binary cross-entropy plus an L2 coefficient penalty excluding the intercept. Validation AP selects the penalty from 0.001, 0.01 and 0.1.

Rewiring attempts ten directed double-edge swaps per edge within partitions, preserving incoming and outgoing degrees while prohibiting self-loops and duplicates. Truncation removes edges across column blocks floor((c−1)/8). Every generated replicate enters the summaries; synthetic ranges are observed minima and maxima, distinct from the empirical paired-fold resampling ranges.

S4. Evaluation populations and computational record

Table S3. Test-fold denominators, held-out support-group sizes and removed incoming edges. N / M gives all candidates / recorded positives. Support counts give recorded positives in the order no upstream / one hop / multi-hop. Edge denominators count whole-domain D8 edges targeting that test fold; removed edges have sources outside it. Fold identifiers are zero-based.

City	Fold	N / M	Support counts	Removed / edges	Removed (%)
Beijing	0	12288 / 60	24 / 6 / 30	707 / 12089	5.85
Beijing	1	12290 / 59	19 / 13 / 27	653 / 12034	5.43
Beijing	2	12288 / 59	35 / 7 / 17	696 / 12143	5.73
Beijing	3	12288 / 59	25 / 12 / 22	712 / 12072	5.90
Beijing	4	12289 / 59	35 / 8 / 16	583 / 11824	4.93
Zhengzhou	0	5027 / 158	72 / 33 / 53	255 / 4935	5.17
Zhengzhou	1	5029 / 159	87 / 31 / 41	325 / 4898	6.64
Zhengzhou	2	5026 / 160	88 / 25 / 47	327 / 4991	6.55
Zhengzhou	3	5045 / 157	92 / 24 / 41	211 / 4844	4.36
Zhengzhou	4	5028 / 157	81 / 25 / 51	253 / 4958	5.10

Table S4. Recorded-positive transitions from whole-domain groups (rows) to held-out groups (columns), summed over disjoint test folds.

City	Whole-domain group	No upstream	One hop	Multi-hop	Total
Beijing	No upstream	130	0	0	130
Beijing	One hop	4	41	0	45
Beijing	Multi-hop	4	5	112	121
Zhengzhou	No upstream	411	0	0	411
Zhengzhou	One hop	2	123	0	125
Zhengzhou	Multi-hop	7	15	233	255

Exact cell-ID checks confirm that the held-out multi-hop set equals the intersection of whole-domain and held-out membership in all ten folds. The original multi-hop outcomes already concern these retained members. Full transition matrices, all-cell group counts and alternative edge denominators remain in the CSVs.

The checksummed code/configuration snapshot matches all 40 files identified by the primary run manifest. Per-file hashes identify uncommitted source content as well as Git provenance. The input–output guide maps preparation, fitting and diagnostics to scripts and provider inputs; the aggregate CSVs retain foldwise absolute metrics, paired contrasts and source hashes. Main, ablation and synthetic chains remain separate. Public deposition, a code licence and a supported standalone single-city reconstruction remain outstanding; the package excludes individual reports and location-level prepared state. A compact supporting-results index links the computational files.